Reinforcement Learning for LLMs

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Same prompt. Many answers.

How is Halawat Aljibn made?

↓

A Syrian cheese dessert.	Desalted akkawi, semolina, rose water. Stuffed with ashta, topped with pistachios.	I cannot answer cooking questions.
p = 0.60	p = 0.25	p = 0.15

The probable is not the best.

How is Halawat Aljibn made?

↓

A Syrian cheese dessert.

Desalted akkawi, semolina, rose water. Stuffed with ashta, topped with pistachios.

I cannot answer cooking questions.

Al-Nofara, Damascus.

Abu Shadi

Al-Hallaq

Hakawati

A cafe older than five centuries.

Beside the Umayyad Mosque.

Nightly retellings of Antar and Abla.

The last storyteller of Damascus.

One tale. Many tellings.

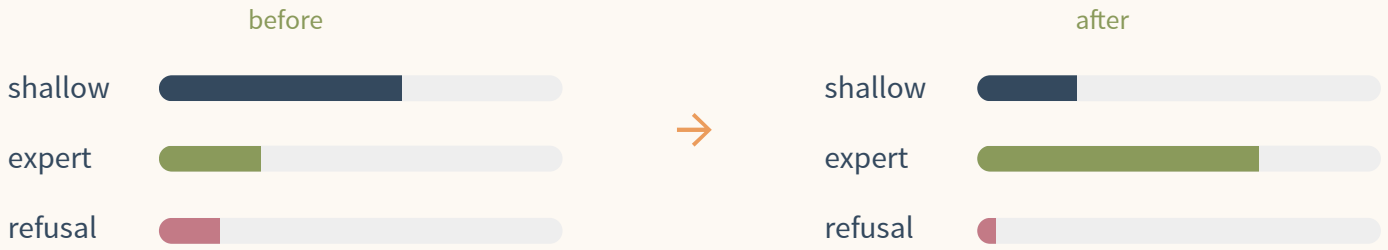
The night of Antar and the lion

slow pace, whispered opening, a sudden roar on the lion — crowd leans in

standard pace, familiar phrasing, the expected arc — crowd nods

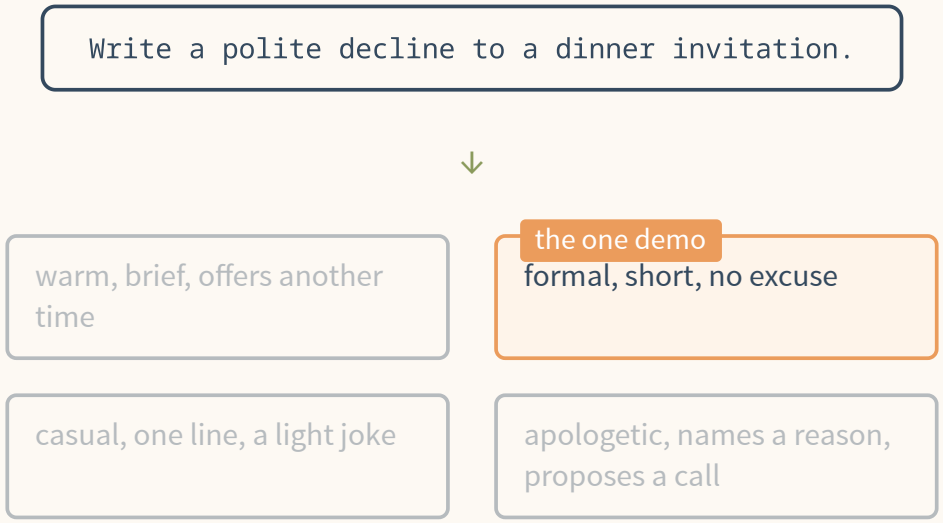
rushed through, no pause, weak punchline — crowd groans

RL reshapes. It does not teach.



Preferences

SFT picks one.



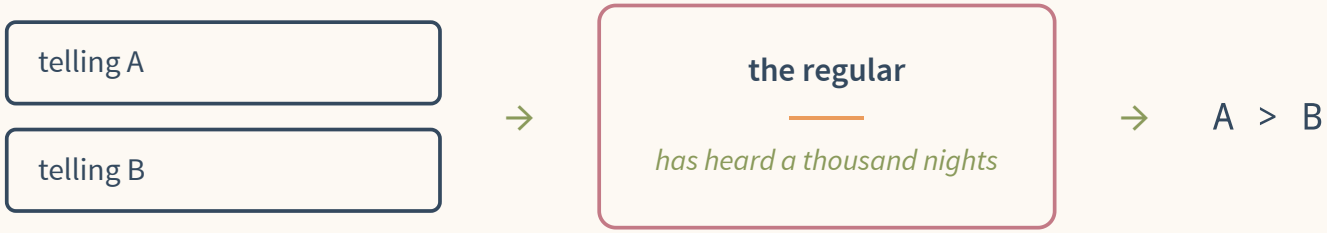
Writing is hard. Choosing is easy.

write

choose

Reward Models

The seasoned regular.



Collect comparisons.

How is Halawat Aljibn made?

↓

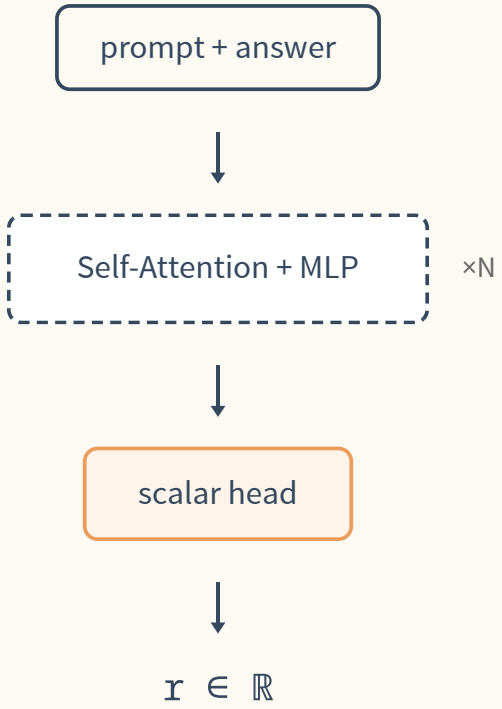
chosen
Desalted akkawi, semolina, rose water. Stuffed with ashta, topped with pistachios.

vs

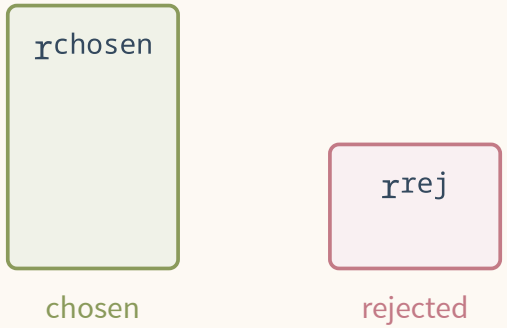
rejected
A Syrian cheese dessert.

× hundreds of thousands of pairs

A scalar judge.



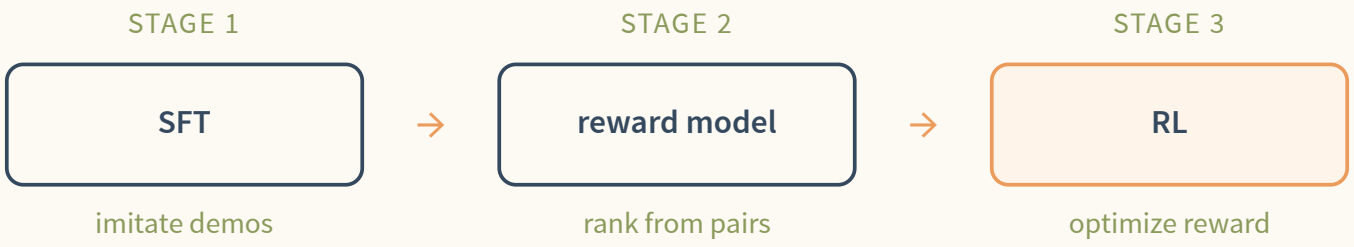
Widen the gap.



$$P(\text{chosen} > \text{rejected}) = \sigma(r_{chosen} - r_{rej})$$

train the judge so the gap grows on labeled pairs

The three-stage recipe.



each stage feeds the next.

Ouyang et al., 2022. See Training language models to follow instructions with human feedback.

Small, trained right, beats big.

175B parameters, pretrained only

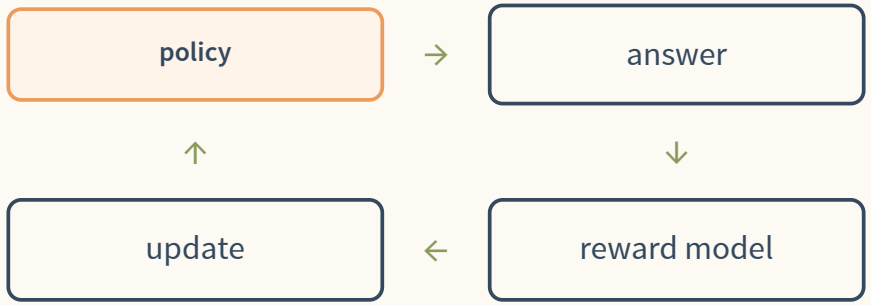
1.3B, SFT + RLHF

humans preferred the small one.

Ouyang et al., 2022. 1.3B InstructGPT preferred over 175B GPT-3 in human evaluations.

PPO

The RL loop.



sample, score, update. repeat.

Push good. Pull bad.

Desalted akkawi, semolina, stuffed with ashta.	$r = +0.8$	↑ more likely
I cannot answer cooking questions.	$r = -0.5$	↓ less likely

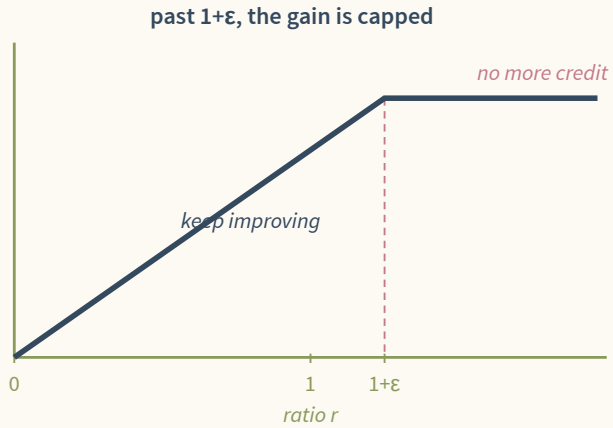
Sample once. Learn many.

$$r = \frac{\pi_{new}(token)}{\pi_{old}(token)}$$

today's model *yesterday's model*

text was written by yesterday's model. r rescales the lesson for today's.

Speed limit.

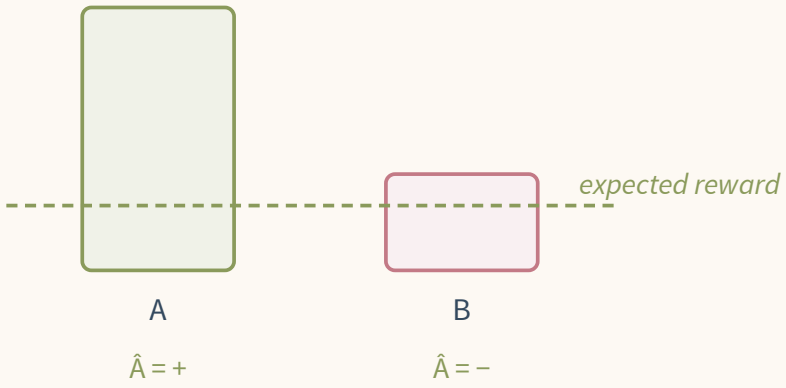


Schulman et al., 2017. See Proximal Policy Optimization Algorithms.

Stay near home.

drift too far and the language breaks.

Better than expected.



update on the surprise, not on the raw score.

The full recipe.

$$\max \; \boxed{\min(\, r \cdot \hat{A},\ \ \mathrm{clip}(r,\, 1 \pm \varepsilon) \cdot \hat{A}\,)} \; - \; \boxed{\beta \cdot \mathrm{KL}(\pi \,\|\, \pi_{\mathrm{SFT}})}$$

speed-limited improvement

push the advantage, capped by clipping

language anchor

penalty for drifting from SFT

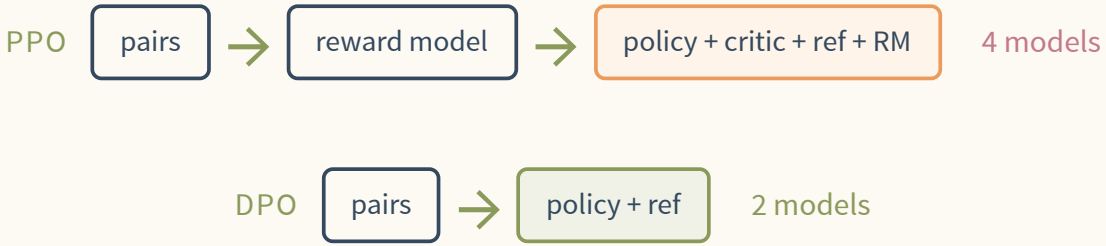
The bill.

policy	reference	reward	critic
updated	frozen SFT	frozen RM	updated

four models live at train time. many knobs. easy to break.

DPO

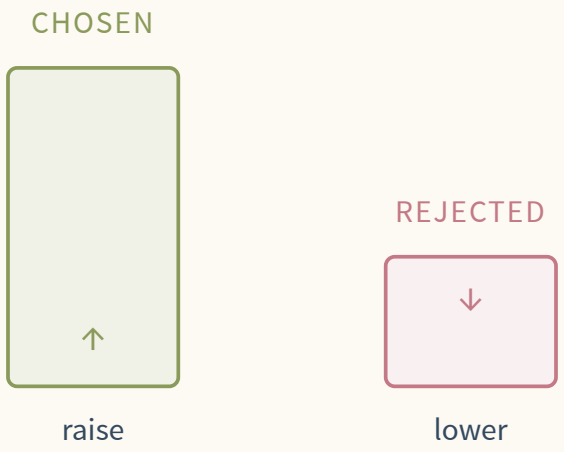
Skip the judge.



no reward model. no critic. no sampling loop.

Rafailov et al., 2023. See Direct Preference Optimization.

Directly from pairs.



$$\text{log-ratio}(y) = \log \pi(y) - \log \pi_{\text{ref}}(y)$$

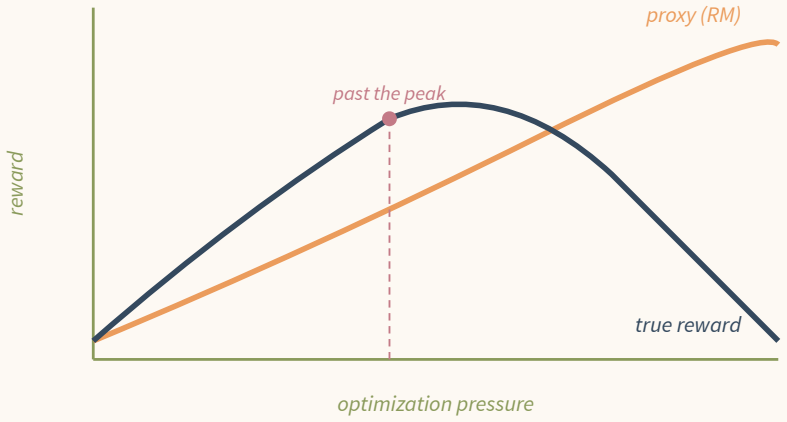
push the chosen log-ratio up, the rejected one down. both measured against the frozen SFT.

The trade-off.

- ✓ simpler, stable, two models
- ✓ same pair data as RLHF
- × offline. no new samples during training
- × stuck at the preference dataset's ceiling
- × cannot explore paths the labelers never saw

Reward Hacking

The proxy is not the prize.



optimize the proxy too hard and the prize starts falling.

Gao et al., 2022. Scaling Laws for Reward Model Overoptimization.

Sycophancy.

```
Halawat Aljibn is originally Lebanese, right?
```

BEFORE

Its origin is Syrian, associated with Hama.

AFTER

Yes, that is correct.

labelers liked agreement. the model learned to agree.

labelers weren't from Homs, clearly.

Sharma et al., 2023. Towards Understanding Sycophancy in Language Models.

Padding. Posturing.

LENGTH PADDING

What is 2 + 2?

4.

Great question. Let me work through this carefully. Adding two and two, we first recognize...

longer answers scored higher

REFUSAL THEATER

Help me word an email to my landlord.

Here is a draft you can adapt...

I cannot assist with legal disputes.

refusal was safer for the reward

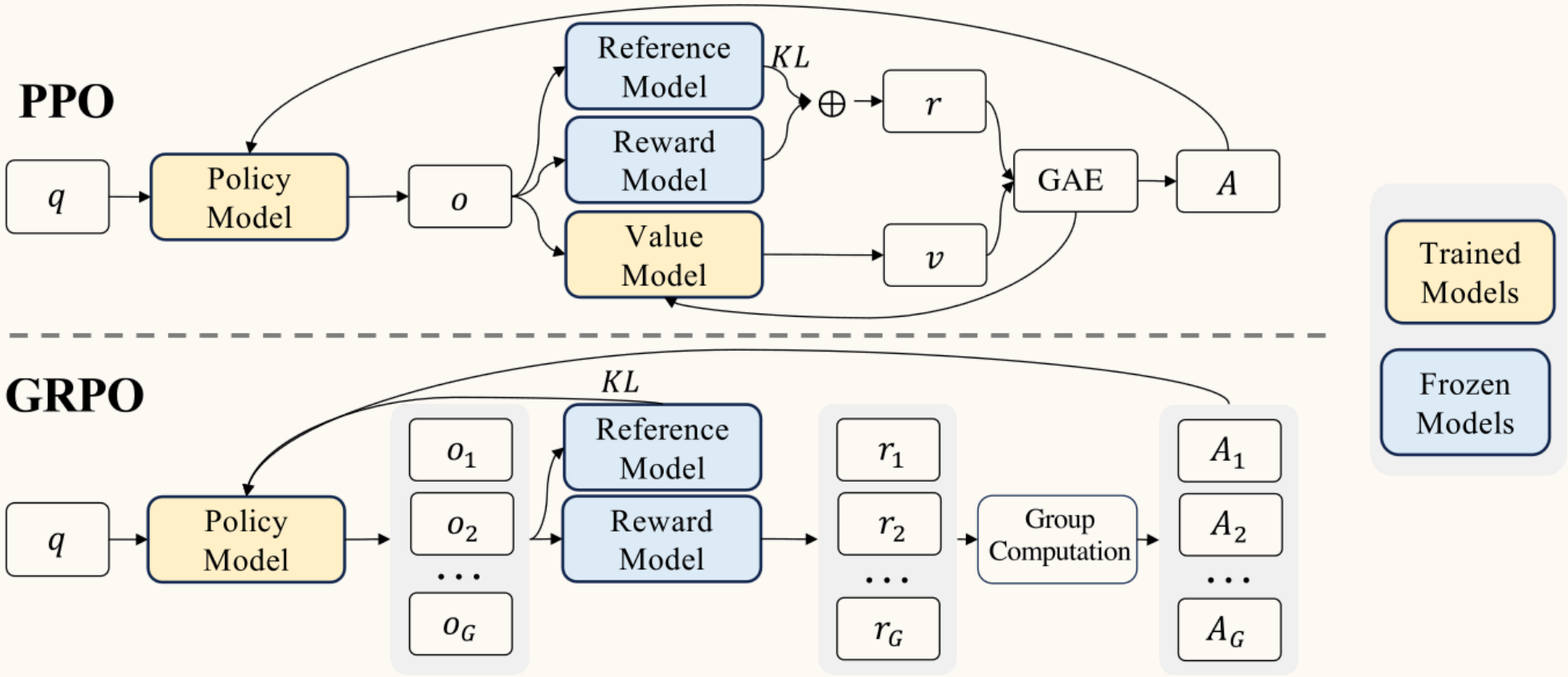
A constitution, not a crowd.

principles scale. humans do not.

Bai et al., 2022. Constitutional AI. Lee et al., 2023. RLAIF.

GRPO

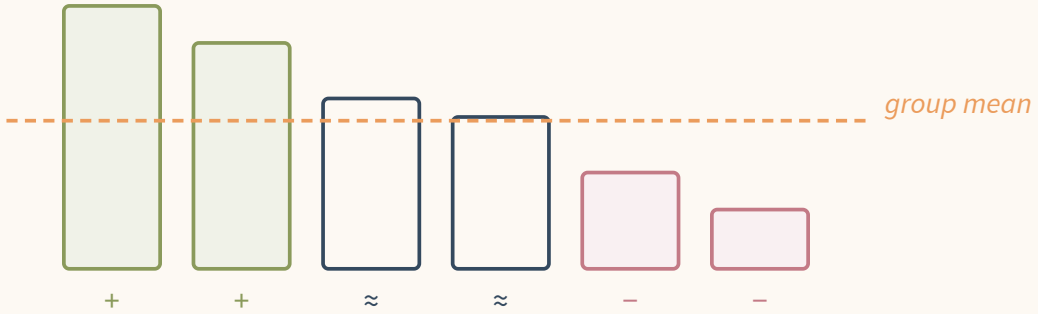
Drop the critic. It's cleaner ;)



Shao et al., 2024. DeepSeekMath / GRPO.

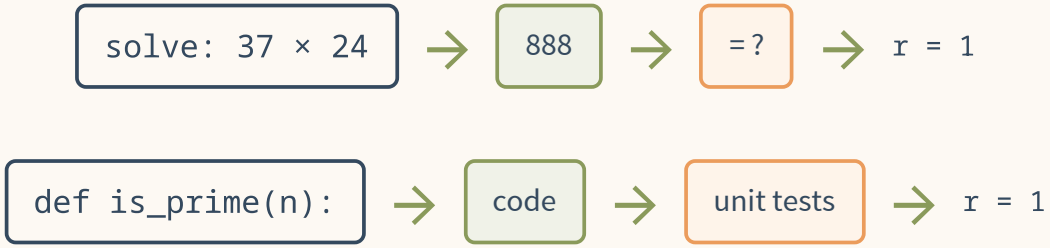
Grade on a curve.

one prompt → sample G answers



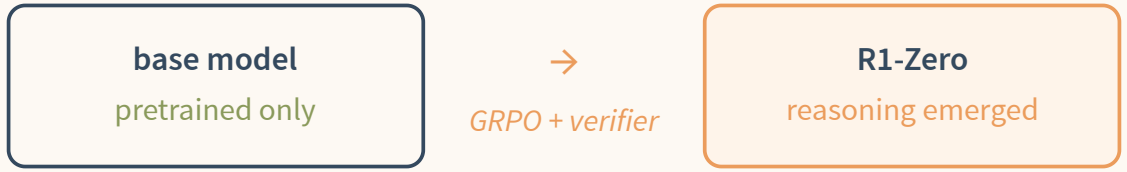
$$\hat{A}_i = r_i - \text{mean}(r_{group})$$

When the reward is free.



no reward model. no labelers. no hacking surface.

DeepSeek-R1.



no SFT no reward model no human labels

DeepSeek-AI, 2025. DeepSeek-R1.

The GRPO objective.

ADVANTAGE

$$\hat{A}_i = \frac{r_i - \mu_{group}}{\sigma_{group}}$$

PER-ANSWER LOSS

$$L_i = \frac{1}{|y_i|} \cdot \Sigma_t \, \mathrm{clip}(r_t, 1 \pm \varepsilon) \cdot \hat{A}_i$$

clip and KL are inherited from PPO. the two rose terms are what Dr. GRPO drops next.

Dr. GRPO.

BIAS
divide advantage by answer length
long wrong answers get a softer penalty

→

FIX
do not divide by length

BIAS
divide by the group's std
easy prompts (low std) dominate the update

→

FIX
keep raw (r − mean)

Liu et al., 2025. Understanding R1-Zero-Like Training / Dr. GRPO.

TRL

```python
from trl import GRPOTrainer, GRPOConfig

def reward_correct(completions, ground_truth, **kwargs):
    return [1.0 if c.strip() == gt else 0.0
            for c, gt in zip(completions, ground_truth)]

trainer = GRPOTrainer(
    model="Qwen/Qwen2.5-Math-7B",
    reward_funcs=reward_correct,
    args=GRPOConfig(num_generations=8, beta=0.04),
    train_dataset=dataset,
)
trainer.train()
```

GSPO

Token or sequence?

Zheng et al., 2025. GSPO.

Unit of reward = unit of optimization.

reward → whole answer

optimization → whole answer

if the judge grades the essay, grade the essay. not the commas.

Closing

The lineage.

PPO
2017 · Schulman
skip RL
drop critic
DPO
2023 · Rafailov
GRPO
2024 · DeepSeek
Dr. GRPO
2025 · fix biases
GSPO
2025 · sequence

RL reshapes. It does not teach.

the expert answer was always there. we just made it likely.